

SAM MOLUGU

Java Backend Engineer · Distributed Systems · FinTech & Healthcare

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Summary

Backend engineer with 5+ years building high-throughput Java systems in FinTech and healthcare. Led event-driven transaction platform at J.P. Morgan Chase processing 50M+ events/day on Kafka, and shipped production systems at Bank of America and CareFirst under strict compliance requirements. Currently building and deploying full-stack SaaS products independently, with end-to-end ownership from architecture to production.

Skills

Languages: Java, Python, JavaScript (Node.js), SQL

Frameworks: Spring Boot, Express.js, Hibernate, REST APIs

Cloud & Infra: AWS (EC2, S3, Lambda, IAM, CloudWatch), Docker, Kubernetes, Terraform, Ansible, Nginx, DigitalOcean

Databases: PostgreSQL, MySQL, Oracle, MongoDB, SQLite, AWS RDS

Messaging: Apache Kafka, AWS SQS

CI/CD: Jenkins, GitHub Actions, GitLab CI, Maven

Observability: Splunk, Datadog, CloudWatch

Testing: JUnit, Mockito, Playwright (E2E), Postman, Swagger

Automation: Python (Typer, Pydantic, httpx, BeautifulSoup, pdfplumber), PyYAML, SQLite

Experience

Freelance Software Engineer | [The Hide Hostel @ Canggu](#) — Bali, Indonesia | Aug 2024 – Present

- Built and deployed a full-stack hospitality management platform for a live Canggu hostel — Node.js 20 + Express REST API, SQLite persistence, JWT auth, Helmet security stack, rate limiting, and HTTPS enforcement via Nginx + Certbot on DigitalOcean.
- Integrated tesseract.js OCR + MRZ parsing for automated passport scanning at guest check-in, eliminating manual data entry for ID verification.
- Built CSV ingestion, XLSX report generation, and Multer file upload workflows unifying booking, payment, and guest data pipelines; delivered PWA with Tailwind frontend and Playwright E2E test suite.
- Containerized with Docker + Docker Compose; deployed to Ubuntu 24.04 droplet with systemd service management for reliable uptime.

Software Engineer II | [J.P. Morgan Chase & Co.](#) — Wilmington, DE | Aug 2023 – Aug 2024

- Led redesign of a card-processing system using Java, Spring Boot, and Kafka with retry + DLQ handling — supporting 50M+ failed transaction events/day in production.
- Partnered on Chase–Amazon SplitPay launch supporting ~\$300M in daily consumer financing volume, improving payment processing reliability across lending operations.
- Built internal log analysis tooling to classify transaction failures across distributed event streams, improving detection of recurring error patterns at scale.

Software Engineer | [Bank of America](#) — Charlotte, NC | Feb 2022 – Mar 2023

- Built a distributed credit-risk evaluation platform enabling real-time scoring and analytics across multi-region financial systems.
- Improved CI quality gates across 60% of the risk-rating codebase by integrating JaCoCo and Mockito metrics into SonarQube, increasing test coverage visibility by ~35%.
- Developed macroeconomic-factor-based credit scoring features via SQL-driven pipelines, with end-to-end QA validation of risk workflows.

Associate Software Engineer | [CareFirst](#) — Washington, DC | Jun 2020 – Feb 2022

- Built HIPAA-compliant Java REST APIs processing 2M+ healthcare claims/day for U.S. federal employee healthcare systems.
- Developed ETL and validation pipelines for healthcare claims data, improving reporting accuracy and reducing reconciliation overhead.

Projects

Job Search Automation Tool | [Personal Project](#) — | 2025

- Built a Python CLI pipeline (Typer, Pydantic, httpx) that ingests a resume PDF via pdfplumber, constructs a structured candidate profile, and queries Greenhouse and Lever ATS APIs to discover and score job matches across company boards — with multi-factor scoring (relevance, trust, freshness) and YAML-driven board configuration with daily rotation.
- Automated form pre-fill and application workflow using Playwright with per-ATS playbooks, a rule-based answer bank with learned patterns, and human-in-the-loop handling for CAPTCHA/submit; unattended fallback mode for required-field failures.
- Audit trail persisted in SQLite (applications, scores, form questions); generates markdown application reports; scheduled via Windows Task Scheduler to run during US Eastern market hours.

Education

Bachelor of Information Technology — **George Mason University** | Fairfax, VA · 2020